

WILLIAM McNULTY

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Software • AI Systems • Data & Analytics • Consulting • Economics

EDUCATION

University of Virginia – B.A. Computer Science | B.A. Economics | Expected May 2028

GPA: 3.6/4.0 | 117 Earned Credits | Industrial Organization Concentration | Real Estate Minor

Relevant Coursework: Software Development Essentials, Data Structures & Algorithms, Computer Systems & Organization, Industrial Organization, Statistics, Theory of Financial Markets

In Progress (Fall 2026): Data Structures & Algorithms 2, Intro to Cybersecurity, Usability Engineering, Applied Linear Algebra, Intermediate Macroeconomics

PROJECTS

UVA Knowledge Network – Local AI Pipeline | Builder | June 2026 – Present

- Building a private, locally hosted UVA knowledge network: converting 700+ course documents and class-recording transcripts across 10 courses into structured, searchable markdown with open-source models on local hardware, and developing retrieval agents that answer questions across courses.
- Developed an image-classification and figure-description workflow with local vision-language models that classifies course pages as graph, diagram, table, or text; tuned prompts and image resolution against a hand-checked evaluation set to reach 100% page-type accuracy.
- Built automated experiment loops that benchmark model, prompt, and resolution variants against the evaluation set and adopt winning configurations, plus a BM25 search prototype over 7,000+ text chunks as the retrieval foundation for cross-course question answering.

Job-Application Tracker | Builder | September 2026 – Present

- Built a local dashboard with a standard-library Python API and a single-page JavaScript front end that persists every change to version-controlled JSON, with automated unit tests covering input validation, persistence, and request-host checks.

Fraternity Operations Platform – AI-Enabled | Product & Operations Lead | June 2026 – Present

- Built a unified chat platform and membership database to replace communication scattered across Flare, Snapchat, and GroupMe, as part of an AI-enabled operations platform that streamlines organizational workflows, reporting, and approvals.
- Drove adoption by addressing member concerns about cost, time to adoption, and design through requirements gathering and iterative releases; other brothers now contribute to the project and it supports alumni connections.

EXPERIENCE

Teaching Assistant | CS 3140 Software Development Essentials, University of Virginia | August 2026 – Present

- Teach course content to a body of ~200 students, translating software development concepts into practical guidance.
- Conduct systems design reviews and programming reviews, giving students technical feedback on their designs and code.

Associate Consultant | Aegis Cyber Solutions | March 2023 – Present

- Evaluated emerging AI technologies and conducted comparative analyses, translating technical findings into strategic recommendations that supported organizational decision-making and technology adoption.
- Prepared executive-facing reports, presentations, and implementation roadmaps while communicating technology strategies and research findings to technical and non-technical stakeholders.
- Leveraged AI-assisted research, analysis, and documentation tools to improve deliverables and accelerate project execution.

Undergraduate Researcher | University of Virginia | Fall 2025

- Collaborated with University of Virginia faculty and World Bank researchers on quantitative research initiatives supporting evidence-based policy analysis.
- Collected, validated, analyzed, and synthesized datasets into reports and presentations for academic stakeholders.

LEADERSHIP

Security Manager | Boylan Heights | March 2025 – Present

- Lead daily operations for a 40+ member organization by coordinating staffing, scheduling, communication, and accountability.
- Develop operational procedures that improve organizational effectiveness and incident response in a fast-paced environment.

SKILLS

Technical: Python • SQL • PostgreSQL • FastAPI • JavaScript • Docker • Git • Unit Testing • AI Tools: Claude Code, Ollama / Local LLMs

Business: Requirements Gathering • Stakeholder Engagement • Data Analysis • Technical Communication • Project Management